1.5 Solution of Linear Equations

As seen from the previous section, the global system of equations after boundary conditions consists of n equations in n unknowns

$$K\,d = R$$

Most scientific calculators have functions for solving linear system of equations. These are useful for solving small textbook problems. Computer algebra systems (*Mathematica*, Matlab, Maple, etc.) have sophisticated built-in functions for solving large system of linear equations. For example using *Mathematica* a system of equations can be solved by first defining matrix K and right hand side vector R and then issuing the following command.

```
LinearSolve[K, R]
```

For completeness two basic methods for solution of large scale linear system of equations are briefly discussed in this section. Detailed treatment of these methods is beyond the scope of this text.

■ 1.5.1 Solution Using Choleski Decomposition

An efficient method to solve the system of equations is to first decompose the matrix K into product of lower and upper triangular matrices. For symmetric matrices the so-called Choleski decomposition is very efficient. In this decomposition a symmetric matrix K is decomposed into a product of a lower triangular matrix L and its transpose L^T in the following form.

$$K = L\,L^T$$

$$L = \begin{pmatrix} \ell_{11} & 0 & 0 & \dots & 0 \\ \ell_{21} & \ell_{22} & 0 & \dots & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ \ell_{n1} & \ell_{n2} & \ell_{n3} & \dots & \ell_{nn} \end{pmatrix}; \quad L^T = \begin{pmatrix} \ell_{11} & \ell_{21} & \ell_{31} & \dots & \ell_{n1} \\ 0 & \ell_{22} & \ell_{32} & \dots & \ell_{n2} \\ 0 & 0 & \ell_{33} & \dots & \ell_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \ell_{nn} \end{pmatrix}$$

Such a decomposition is always possible for nonsingular symmetric matrices. Assuming this decomposition is known, we can solve the system of equations in two steps as follows.

$$K\,d = R \implies L\,L^T\,d = R$$

If we define $L^T\,d = y$ then

$$L\,L^T\,d = R \implies L\,y = R \implies \begin{pmatrix} \ell_{11} & 0 & 0 & \dots & 0 \\ \ell_{21} & \ell_{22} & 0 & \dots & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ \ell_{n1} & \ell_{n2} & \ell_{n3} & \dots & \ell_{nn} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ \vdots \\ r_n \end{pmatrix}$$

Since L is a lower triangular matrix, the first row gives the solution for the first variable as $y_1 = r_1 / \ell_{11}$. Knowing this, we can solve for $y_2 = (r_2 - \ell_{21}\, y_1)/\ell_{22}$ from the second equation. Thus proceeding forward from the first equation we can solve for all intermediate variables y_i. This is known as *forward elimination*.

$$y_i = (r_i - \sum_{k=1}^{i-1} \ell_{ik}\, y_k)/\ell_{ii};\quad i = 1, 2, \ldots, n$$

Now we can solve for the actual unknowns d_i as follows.

$$L^T d = y \implies \begin{pmatrix} \ell_{11} & \ell_{21} & \ell_{31} & \ldots & \ell_{n1} \\ 0 & \ell_{22} & \ell_{32} & \ldots & \ell_{n2} \\ 0 & 0 & \ell_{33} & \ldots & \ell_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & \ell_{nn} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix}$$

In this system the last equation has only one unknown giving $d_n = y_n / \ell_{nn}$. The second to last equation is used next to get d_{n-1} and thus working backward from the last equation we can solve for all unknowns. This is known as *backward substitution*.

$$d_i = (y_i - \sum_{k=i+1}^{n} \ell_{ki}\, d_k)/\ell_{ii};\quad i = n, n-1, \ldots, 1$$

The procedure is especially efficient if solutions for several different right hand sides are needed. Since in this case the major computational task of decomposing the matrix into $L\,L^T$ form is not necessary for every right hand side. One simply needs to perform a forward elimination and a backward substitution for each different right hand side.

Choleski decomposition

The Choleski decomposition algorithm can be developed by considering direct multiplication of terms in the L and L^T matrices and equating them to the corresponding terms in the K matrix.

$$K = L\,L^T \implies \begin{pmatrix} k_{11} & k_{21} & k_{31} & \ldots & k_{n1} \\ k_{21} & k_{22} & k_{32} & \ldots & k_{n2} \\ k_{31} & k_{32} & k_{33} & \ldots & k_{n3} \\ . & . & . & . & . \\ k_{n1} & k_{n2} & k_{n3} & \ldots & k_{nn} \end{pmatrix} = \begin{pmatrix} \ell_{11} & 0 & 0 & \ldots & 0 \\ \ell_{21} & \ell_{22} & 0 & \ldots & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} & \ldots & 0 \\ . & . & . & . & . \\ \ell_{n1} & \ell_{n2} & \ell_{n3} & \ldots & \ell_{nn} \end{pmatrix} \begin{pmatrix} \ell_{11} & \ell_{21} & \ell_{31} & \ldots & \ell_{n1} \\ 0 & \ell_{22} & \ell_{32} & \ldots & \ell_{n2} \\ 0 & 0 & \ell_{33} & \ldots & \ell_{n2} \\ . & . & . & . & . \\ 0 & 0 & 0 & \ldots & \ell_{nn} \end{pmatrix}$$

By direct multiplication we can see that the entries in the first row of L matrix can be established as follows.

Row 1: $\quad k_{11} = \ell_{11}^2 \implies \ell_{11} = \sqrt{k_{11}}$

$$k_{21} = \ell_{11}\,\ell_{21} \implies \ell_{21} = k_{21}/\ell_{11}$$

$$k_{i1} = \ell_{11}\,\ell_{i1} \implies \ell_{i1} = k_{i1}/\ell_{11};\quad i = 2, \ldots, n$$

Once entries in row 1 are known, we can proceed to row 2 and establish the entries there by direct multiplication as follows.

Row 2:

$$k_{22} = \ell_{21}^2 + \ell_{22}^2 \implies \ell_{22} = \sqrt{k_{22} - \ell_{21}^2}$$

$$k_{32} = \ell_{21}\,\ell_{31} + \ell_{22}\,\ell_{32} \implies \ell_{32} = (k_{32} - \ell_{21}\,\ell_{31})/\ell_{22}$$

$$k_{i2} = \ell_{21}\,\ell_{i1} + \ell_{22}\,\ell_{i2} \implies \ell_{i2} = (k_{i2} - \ell_{21}\,\ell_{i1})/\ell_{22};\quad i = 3, \ldots, n$$

Proceeding in a similar manner, we can write the following general formulas for entries in the ith row of the L matrix.

Row i:

$$k_{ii} = \ell_{i1}^2 + \ell_{i2}^2 + \ldots + \ell_{ii}^2 \implies \ell_{ii} = \sqrt{k_{ii} - \sum_{k=1}^{i-1} \ell_{ik}^2}$$

$$k_{ij} = \ell_{i1}\,\ell_{j1} + \ell_{i2}\,\ell_{j2} + \ldots + \ell_{ij}\,\ell_{jj} \implies \ell_{ij} = \left(k_{ij} - \sum_{k=1}^{j-1} \ell_{ik}\,\ell_{jk}\right)/\ell_{jj}$$

Notice that the procedure requires taking square root of terms coming from the diagonal of the K matrix and then division by this term to establish other terms in a given row. This is called the pivot element. Clearly these operations will fail if there is a negative or zero pivot. However as long as the matrix K is non-singular it is always possible to re-arrange the system of equations to avoid negative or a zero pivot.

The computational steps can be written in the form of the following algorithm.

Choleski decomposition algorithm

For $i = 1, n$

For $j = 1, i - 1$

$$\ell_{ij} = \left(k_{ij} - \sum_{k=1}^{j-1} \ell_{ik}\,\ell_{jk}\right)/\ell_{jj}$$

end

$$\ell_{ii} = \sqrt{k_{ii} - \sum_{k=1}^{i-1} \ell_{ik}^2}$$

end

Example 1.13

Find solution of the following system of equations using Choleski decomposition

$$\begin{pmatrix} 9 & 3 & 6 & 3 \\ 3 & 12 & 2 & -10 \\ 6 & 2 & 11 & -5 \\ 3 & -10 & -5 & 21 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 7 \\ 8 \end{pmatrix}$$

We first generate $L\,L^T$ decomposition of the coefficient matrix using the above algorithm.

$$K = LL^T = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 1 & \sqrt{11} & 0 & 0 \\ 2 & 0 & \sqrt{7} & 0 \\ 1 & -\sqrt{11} & -\sqrt{7} & \sqrt{2} \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 & 1 \\ 0 & \sqrt{11} & 0 & -\sqrt{11} \\ 0 & 0 & \sqrt{7} & -\sqrt{7} \\ 0 & 0 & 0 & \sqrt{2} \end{pmatrix} = \begin{pmatrix} 9 & 3 & 6 & 3 \\ 3 & 12 & 2 & -10 \\ 6 & 2 & 11 & -5 \\ 3 & -10 & -5 & 21 \end{pmatrix}$$

Using the L matrix the solution for intermediate variables (forward elimination) is as follows.

$$\begin{pmatrix} 3 & 0 & 0 & 0 \\ 1 & \sqrt{11} & 0 & 0 \\ 2 & 0 & \sqrt{7} & 0 \\ 1 & -\sqrt{11} & -\sqrt{7} & \sqrt{2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 7 \\ 8 \end{pmatrix}$$

Row 1: $3\,y_1 = 5 \Longrightarrow y_1 = \dfrac{5}{3}$

Row 2: $1\,y_1 + \sqrt{11}\,y_2 = 6 \Longrightarrow y_2 = \dfrac{13}{3\sqrt{11}}$

Row 3: $2\,y_1 + 0\,y_2 + \sqrt{7}\,y_3 = 7 \Longrightarrow y_3 = \dfrac{11}{3\sqrt{7}}$

Row 4: $1\,y_1 + (-\sqrt{11})\,y_2 + (-\sqrt{7})\,y_3 + \sqrt{2}\,y_4 = 8 \Longrightarrow y_4 = \dfrac{43}{3\sqrt{2}}$

Using the L^T matrix the solution for actual variables (backward substitution) is as follows.

$$\begin{pmatrix} 3 & 1 & 2 & 1 \\ 0 & \sqrt{11} & 0 & -\sqrt{11} \\ 0 & 0 & \sqrt{7} & -\sqrt{7} \\ 0 & 0 & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} \\ \frac{13}{3\sqrt{11}} \\ \frac{11}{3\sqrt{7}} \\ \frac{43}{3\sqrt{2}} \end{pmatrix}$$

Row 4: $\sqrt{2}\,d_4 = \dfrac{43}{3\sqrt{2}} \Longrightarrow d_4 = \dfrac{43}{6}$

Row 3: $\sqrt{7}\,d_3 + (-\sqrt{7})\,d_4 = \dfrac{11}{3\sqrt{7}} \Longrightarrow d_3 = \dfrac{323}{42}$

Row 2: $\sqrt{11}\,d_2 + 0\,d_3 + (-\sqrt{11})\,d_4 = \dfrac{13}{3\sqrt{11}} \Longrightarrow d_2 = \dfrac{499}{66}$

Row 1: $3\,d_1 + 1\,d_2 + 2\,d_3 + 1\,d_4 = \dfrac{5}{3} \Longrightarrow d_1 = -\dfrac{730}{77}$

■ 1.5.2 Solution Using Gauss-Seidel Iteration

Gauss–Seidel Iteration

Another general class of methods (other than the elimination methods) used to solve systems of linear algebraic equations is the *iterative methods*. Iterative methods work well when the system of equations is large and sparse (many zero coefficients). The Gauss–Seidel method starts with the original set of equations $[a]\{x\} = \{c\}$ written in the form

$$\begin{aligned} x_1 &= \frac{1}{a_{11}}(c_1 - a_{12}x_2 - a_{13}x_3 - \cdots - a_{1n}x_n) \\ x_2 &= \frac{1}{a_{22}}(c_2 - a_{21}x_1 - a_{23}x_3 - \cdots - a_{2n}x_n) \\ &\vdots \\ x_n &= \frac{1}{a_{nn}}(c_1 - a_{n1}x_1 - a_{n2}x_2 - \cdots - a_{n,n-1}x_{n-1}) \end{aligned} \qquad \text{(B.3.29)}$$

The following steps are then applied.

1. Assume a set of initial values for the unknowns $x_1, x_2, \ldots, x_n$, and substitute them into the right side of the first of Eqs. (B.3.29) to solve for the new x_1.
2. Use the latest value for x_1 obtained from Step 1 and the initial values for $x_3, x_4, \ldots, x_n$ in the right side of the second of Eqs. (B.3.29) to solve for the new x_2.
3. Continue using the latest values of the xs obtained in the left side of Eqs. (B.3.29) as the next trial values in the right side for each succeeding step.
4. Iterate until convergence is satisfactory.

A good initial set of values (guesses) is often $x_i = c_i / a_{ii}$. An example will serve to illustrate the method.

EXAMPLE B.3

Consider the set of linear simultaneous equations given by

$$\begin{aligned} 4x_1 - x_2 &= 2 \\ -x_1 + 4x_2 - x_3 &= 5 \\ -x_2 + 4x_3 - x_4 &= 6 \\ -x_3 + 2x_4 &= -2 \end{aligned} \qquad \text{(B.3.30)}$$

Determine x_1 through x_4.

SOLUTION:

Using the initial guesses given by $x_i = c_i/a_{ii}$, we have

$$x_1 = \frac{2}{4} = \frac{1}{2} \quad x_2 = \frac{5}{4} \approx 1 \quad x_3 = \frac{6}{4} \approx 1 \quad x_4 = -1$$

Solving the first of Eqs. (B.3.30) for x_1 yields

$$x_1 = \frac{1}{4}(2 + x_2) = \frac{1}{4}(2 + 1) = \frac{3}{4}$$

Solving the second of Eqs. (B.3.30) for x_2, we have

$$x_2 = \frac{1}{4}(5 + x_1 + x_3) = \frac{1}{4}(5 + \frac{3}{4} + 1) = 1.68$$

Solving the third of Eqs. (B.3.30) for x_3, we have

$$x_3 = \frac{1}{4}(6 + x_2 + x_4) = \frac{1}{4}[6 + 1.68 + (-1)] = 1.672$$

Solving the fourth of for Eqs. (B.3.30) for x_4, we obtain

$$x_4 = \frac{1}{2}(-2 + x_3) = \frac{1}{2}(-2 + 1.67) = -0.16$$

The first iteration has now been completed. The second iteration yields

$$\begin{aligned} x_1 &= \frac{1}{4}(2 + 1.68) = 0.922 \\ x_2 &= \frac{1}{4}(5 + 0.922 + 1.672) = 1.899 \\ x_3 &= \frac{1}{4}[6 + 1.899 + (-0.16)] = 1.944 \\ x_4 &= \frac{1}{2}(-2 + 1.944) = -0.028 \end{aligned}$$

■ **Table B–1** Results of four iterations of the Gauss–Seidel method for Eqs. (B.3.30)

Iteration	x_1	x_2	x_3	x_4
0	0.5	1.0	1.0	−1.0
1	0.75	1.68	1.672	−0.16
2	0.922	1.899	1.944	−0.028
3	0.975	1.979	1.988	−0.006
4	0.9985	1.9945	1.9983	−0.0008
Exact	1.0	2.0	2.00	0

Table B–1 lists the results of four iterations of the Gauss–Seidel method and the exact solution. From Table B–1, we observe that convergence to the exact solution has proceeded rapidly by the fourth iteration, and the accuracy of the solution is dependent on the number of iterations.